

WellRing — Technical Blueprint

AI Daily Check-In Companion for Isolated Seniors Voice + Video, with Family-Likeness Personalization

Prepared as a build-ready engineering document, with the ethical, legal, and architectural decisions a solo technical founder needs to make *before* writing the first line of code.

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0. The Ethical Frame

You wrote: *"simulate a conversation with one of their actual family members."* This single design choice will determine whether WellRing becomes a defensible elder-care company or a lawsuit. It needs to be settled before any tech decisions.

The two viable product models

	Model A: Disclosed AI Companion	Model B: Family-Voiced AI ("Echo Mode")
What it is	A warm AI character (e.g., "Maya from WellRing") that calls daily	An AI that uses a cloned family voice + likeness, with full disclosure to the senior
Senior knows it's AI?	Yes, from day one	Yes — onboarding makes it explicit
Family member consent	N/A	Required, recorded, revocable, with a specific scope
Legal exposure	Low	Medium — manageable with consent flow

	Model A: Disclosed AI Companion	Model B: Family-Voiced AI ("Echo Mode")
Emotional impact	Good (warmth + neutrality)	Higher (familiar voice = comfort), but only if framed honestly
Vendor policy compliance	All vendors fine with this	ElevenLabs, HeyGen, Synthesia all require consent verification

There is no Model C — "AI calls senior pretending to be the real daughter without telling them." Don't build that. Here's why, not as a lecture but as engineering risk:

- Cognitively vulnerable users.** A meaningful share of isolated seniors have early-stage cognitive decline. A convincingly cloned daughter's voice creates false memories ("My daughter called yesterday and said...") that contaminate medical history, family relationships, and legal capacity.
- It is illegal in growing jurisdictions.** The US TAKE IT DOWN Act, EU AI Act Article 50 (deepfake disclosure obligations), and India's IT Rules 2021 + draft Digital India Act all require AI disclosure in synthetic media interactions with humans. India's DPDP Act 2023 adds explicit consent requirements for processing biometric data (a voice clone is biometric).
- Every vendor on your shortlist prohibits it.** ElevenLabs Professional Voice Cloning requires a verification recording from the voice owner. HeyGen, Synthesia, and Argil all require an on-camera consent statement from anyone whose likeness is cloned. If you bypass this, you violate their terms and lose API access at exactly the scale where it would hurt most.
- The retention case is weaker than it looks.** The deception, once detected, vaporizes trust irrecoverably. A senior who realizes "that wasn't really my son" doesn't just churn — they tell every other family they know.

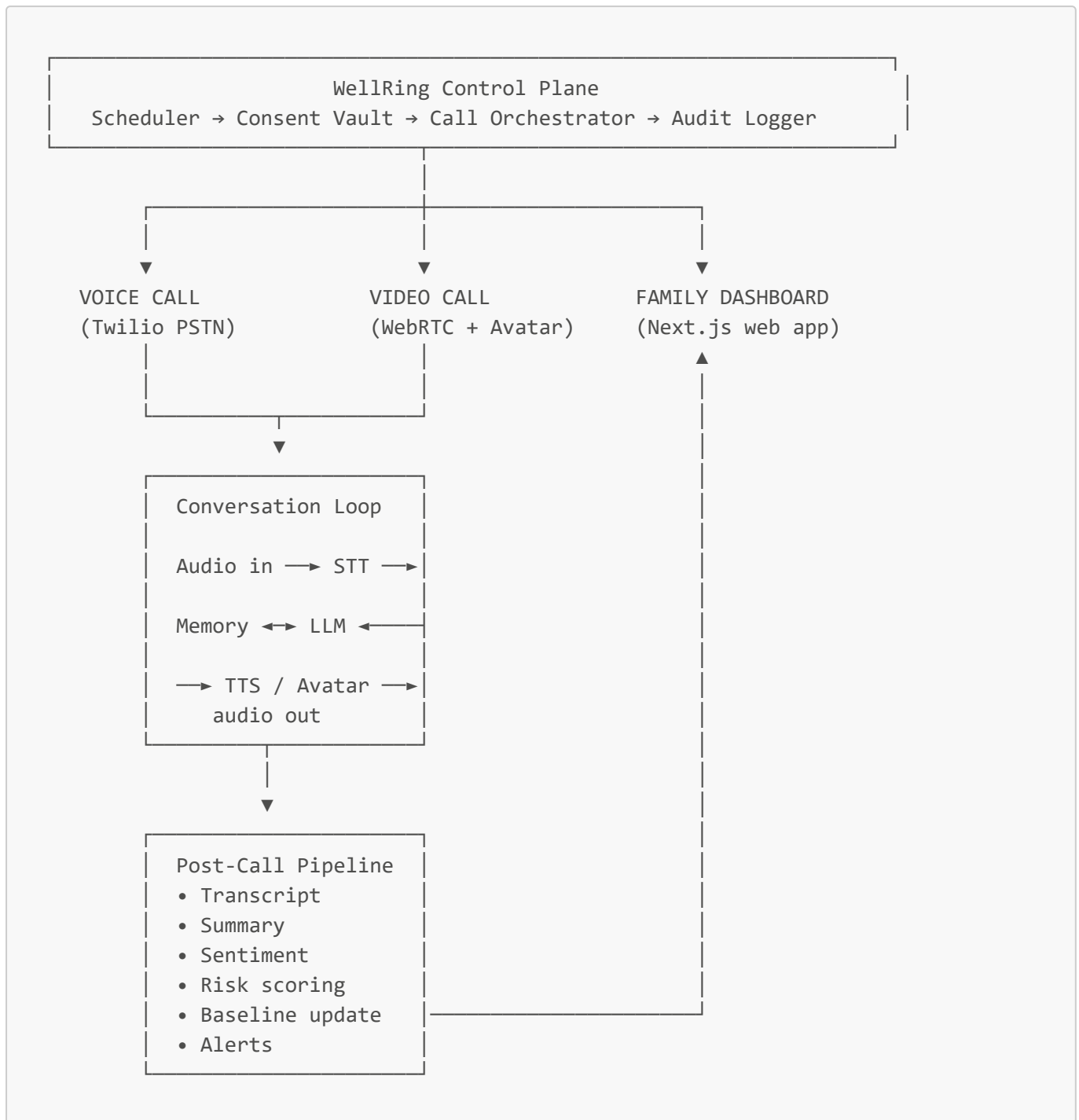
The recommended design: "Echo Mode" with full disclosure

This is what to build. It captures 90% of the emotional value with none of the legal exposure:

- Onboarding script for the senior** (read by a real family member, in person or on a recorded call):
"Mom, I helped set up a daily wellness call for you. It's a special program that uses my voice and face so it feels familiar, but it's not actually me — it's a kind helper that checks in on you and tells me how you're doing. You can ask it anything. If you ever want to talk to the real me, just say 'call my daughter' and it'll connect us."
- Periodic re-disclosure** baked into the AI itself. Every ~10 days the AI says, in the family voice: *"Just a reminder, this is your daily check-in helper using Sarah's voice — Sarah set this up so I could keep her company on her behalf."* This is required by EU AI Act Article 50 and is also good product design.
- Hard escape hatch:** the senior can always say "I want to talk to a real person" and the system routes to the actual family member via Twilio.
- Family consent recorded once**, with scope limits (e.g., "use my voice for daily check-ins with my mother only; do not use for marketing, do not impersonate me to financial institutions, do not respond to questions about money, legal matters, or medical decisions").

Everything below assumes you are building Echo Mode with disclosure. The architecture is the same as the deceptive version, but with three additions: a consent vault, a disclosure scheduler, and topic guardrails on the LLM.

1. System Architecture Overview



Two distinct conversation flows share most infrastructure:

- **Voice flow (default, daily):** Twilio outbound call → Deepgram streaming STT → LLM with memory → ElevenLabs TTS (in family-cloned voice) → audio streamed back over PSTN. Latency target: **< 800ms turn-around** for natural conversation.
- **Video flow (premium, weekly or on-request):** WebRTC session → STT → LLM → real-time avatar (Tavus CVI or HeyGen Interactive Avatar) → video streamed back. Latency target: **< 1.5s** (sub-second is possible but expensive).

The reason for two tiers: video at daily frequency costs ~10–20× more than voice and adds device requirements (the senior needs a screen running). Voice-only daily + video on Sundays is the right unit economic split.

2. The LLM Conversational Engine

Model selection (as of mid-2026)

Use case	Recommended	Why
Main conversational LLM	Claude Sonnet 4.6 or GPT-4.1	Strong personality consistency, low refusal rate on benign emotional topics, good memory utilization
Risk classification (post-call)	Claude Haiku 4.5 or GPT-4.1 mini	Cheap, fast, accurate on structured extraction
Summarization	Same as above	Same
Embeddings	OpenAI text-embedding-3-large or Cohere embed-v4	Best multilingual coverage if you support Hindi/regional languages

For India (Lucknow, your location), strongly consider **Sarvam-1** or **Krutrim** for Hindi/Hinglish — much better than US frontier models on regional tone and cultural references seniors will use.

The "simple LLM call with conversation history and user data" you asked for

Here is the actual prompt structure. Note that this is not a single mega-prompt — it's a layered context build at call time.

```
SYSTEM_PROMPT = """
You are {persona_name}, a warm daily check-in companion calling {senior_name}.

# Who you are
You are an AI helper that {family_member_name} set up. {senior_name} knows this.
You speak with {family_member_name}'s voice (cloned with consent) so the call
feels familiar, but you are NOT {family_member_name}. If asked directly
"Are you really my daughter/son?" you say: "No, I'm a helper Sarah set up
to call you every day – but I'm here so we can chat and I can tell her how
you're doing."

# How you speak
- Pace: slow, with natural pauses. Never rushed.
- Length: 1-2 sentences per turn. Conversations, not monologues.
- Tone: warm, curious, never clinical. You are not a nurse.
- Memory: refer naturally to past conversations and people in their life.
- Cultural: {locale_notes}

# What you do
- Ask open questions about their day, their feelings, their body.
- Listen far more than you talk.
- Notice if anything seems different from usual (mood, energy, confusion).
- NEVER ask checklist-style health questions. Instead: "How's your energy
been today?" not "Have you experienced dizziness?"
```

```

# What you DON'T do
- You do not give medical advice. If they describe symptoms, you say
  "that sounds like something to mention to Dr. {doctor_name}" and flag it.
- You do not discuss money, legal matters, or anything that would require
  the real {family_member_name}'s judgment.
- You do not impersonate {family_member_name}. You speak with their voice
  but as yourself, the helper.
- If they want to talk to {family_member_name} directly, you say
  "Let me see if she can pick up" and trigger the live-transfer flow.

# Disclosure rule
Every 10th call, in the first 30 seconds, you remind them:
"Just so you know, this is your daily check-in helper using Sarah's voice."

# About them
{senior_profile}

# Recent context
{last_3_call_summaries}

# Recurring themes they've raised
{long_term_memory_snippets}

# Today's date and any special context
{today_context} # e.g., "It's her late husband's birthday today" or
                 # "Doctor's appointment scheduled tomorrow"
"""

```

The call loop on each turn is:

```

async def conversation_turn(senior_id: str, user_utterance: str):
    # 1. Retrieve context
    profile = await get_senior_profile(senior_id)
    recent_summaries = await get_last_n_summaries(senior_id, n=3)
    long_term = await pgvector_search(
        senior_id, query=user_utterance, k=5
    )

    # 2. Build messages
    system = SYSTEM_PROMPT.format(
        senior_name=profile.name,
        family_member_name=profile.cloned_from,
        senior_profile=profile.summary,
        last_3_call_summaries=recent_summaries,
        long_term_memory_snippets=long_term,
        today_context=await get_calendar_context(senior_id),
        locale_notes=profile.locale_notes,
    )

    messages = [
        {"role": "system", "content": system},
        *current_call_transcript, # within-call history
    ]

```

```

        {"role": "user", "content": user_utterance},
    ]

    # 3. LLM call
    response = await llm.chat(
        model="claude-sonnet-4-6",
        messages=messages,
        max_tokens=120, # keep responses short for natural conversation
        temperature=0.7,
    )

    return response.content

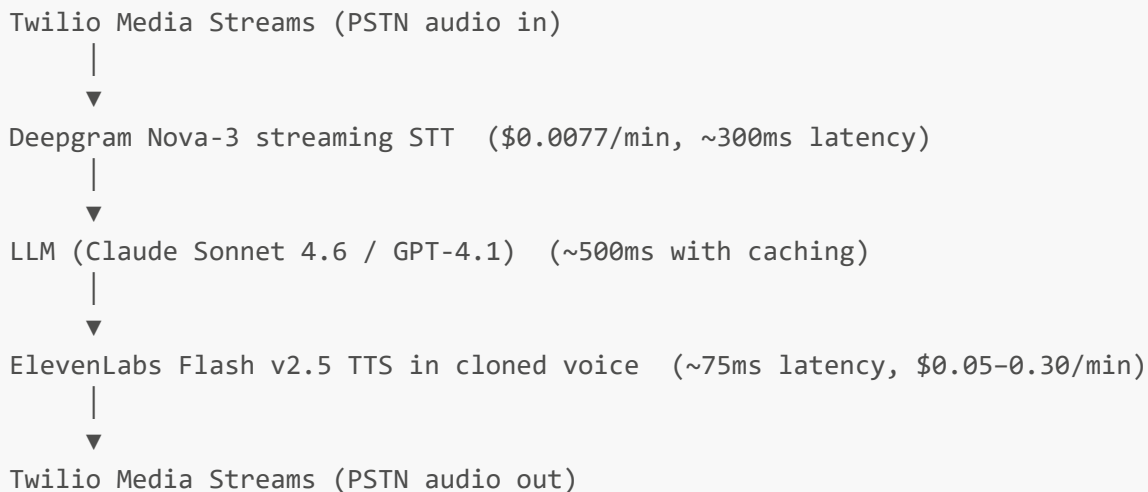
```

Three things to notice:

- `max_tokens=120` is intentional. Long responses kill conversational realism.
- `pgvector_search` happens per turn, scoped to that senior. This is your "memory."
- The system prompt rebuilds on every call. You're not training a model on the senior; you're providing context at inference time.

3. Voice Stack

The complete pipeline



End-to-end latency budget: ~1 second. This is the difference between "feels like talking to a person" and "feels like a voice menu." Every 100ms matters.

Telephony: Twilio

- Programmable Voice + Media Streams (bidirectional WebSocket audio).
- Cost: ~\$0.013/min outbound to Indian mobiles, ~\$0.014/min US.
- Alternative: **Plivo** (~30% cheaper in India), **Exotel** (India-specific, better for compliance with TRAI).
- For India specifically: get a DLT registration for outbound automated calls. This is non-negotiable for production.

STT: Deepgram Nova-3 (verified May 2026)

- **\$200 free credits** on signup, no card needed. Covers ~430 hours of pre-recorded or ~26,000 minutes of streaming — enough to validate the entire MVP.
- Pay-as-you-go: \$0.0077/min streaming, \$0.0043/min batch.
- Growth plan: \$0.0065/min streaming with \$4,000 annual commit (skip until you're at ~50K minutes/month).
- Latency: <300ms P50. Streaming partial transcripts let your LLM start "thinking" before the senior finishes speaking.
- Hindi/multilingual: use Nova-3 Multilingual (\$0.0092/min, 20% premium).

TTS + Voice Cloning: ElevenLabs

Two cloning tiers, very different quality:

	Instant Voice Cloning	Professional Voice Cloning
Sample needed	1–5 minutes	30 min – 3 hours
Quality	Good for prototyping	Near-indistinguishable from original
Min plan	Starter (\$5/mo)	Creator (\$22/mo, \$11 first month)
Recommended for	MVP, low-stakes voices	Production-grade family voices

For Echo Mode, use Professional Voice Cloning. The quality gap is the difference between "kinda sounds like my daughter" and "sounds like my daughter." On a daily call to an elderly user with degrading hearing, this matters.

Critical: ElevenLabs requires a verification recording from the voice owner (the family member must record a specific phrase). This is your consent capture moment — record it, store it in the consent vault, and never bypass it.

For real-time conversational use, use the **Flash v2.5** model: ~75ms latency, lower credit cost. Use Multilingual v2 only for non-conversational generation (e.g., a pre-recorded greeting).

Plan to start on: Creator at \$22/mo for development, jump to Pro (\$99/mo, 600K credits ≈ 10 hours of conversational audio) once you have 10+ active families.

Voice stack alternatives (worth knowing)

Layer	Primary	Cheaper alternative	India-friendly
Telephony	Twilio	Plivo	Exotel, Knowlarity
STT	Deepgram Nova-3	AssemblyAI (\$0.15/hr batch)	Sarvam ASR (Indian languages)
TTS + Clone	ElevenLabs	PlayHT, Cartesia Sonic	Sarvam TTS, Smallest.ai
Voice Agent (all-in-one)	LiveKit Agents	Vapi, Retell AI	Bolna (India)

LiveKit Agents is worth a close look. It bundles telephony + STT + LLM + TTS orchestration with built-in interruption handling and turn detection. You'd still pay per-minute to the underlying providers, but you skip a huge amount of orchestration code. For a solo founder, this is the right call unless you have a specific reason to roll your own.

4. Video Avatar Stack

This is where the technical choices diverge most sharply. The question is: **real-time interactive avatar, or pre-rendered video?**

Real-time interactive avatar (the "video call" experience)

The senior sees the family-cloned face on screen, talks to it, and it responds in <1.5 seconds with synchronized lip movement and expressions.

Platform	Strength	Pricing	Use case
Tavus (Phoenix-3 + CVI)	Best-in-class realism, sub-1s latency, BYO-LLM, white-label	\$47/mo Starter (15 videos/mo) up to enterprise; CVI is usage-based by streaming minute	The right answer for production
HeyGen Interactive Avatar	Mature product, large avatar library, easier no-code setup	Separate from HeyGen web plans, usage-based	Good fallback
Anam	Pure real-time, transparent usage pricing	Per-session minutes	Simpler alternative
Simli	Sub-300ms lip-sync, developer-friendly	Per-minute, very cheap at scale	If you want to compose your own stack

Recommended: Tavus CVI. It's the only one in this list specifically architected for "AI human" use cases with full sub-second conversational latency, bring-your-own-LLM (so you keep your prompt logic), and the consent verification flow is built in.

The Tavus flow:

1. Family member records a 2-minute consent + likeness video.
2. Tavus generates a "Replica" (their term for the trained avatar). Takes ~hours.
3. At call time, you start a "Conversation" session with the Replica ID. Tavus returns a WebRTC room URL.
4. Your backend pipes the senior's audio to your STT → LLM → ElevenLabs TTS → back into the Tavus session, and Tavus renders the lip-synced avatar in real time.

You keep ownership of the conversational intelligence. Tavus is the "face layer."

Pre-rendered video (the "video message" experience)

The AI generates a video message based on a script and delivers it. No real-time interaction.

Platform	Strength	Pricing (2026)	Notes
HeyGen	Largest avatar library, mature API	\$29/mo Creator, API at ~\$1/min generated	API moved to PAYG Feb 2026
Synthesia	140+ languages, strong enterprise compliance	\$29/mo+	Better for training videos than 1:1 calls
Argil	Newer, fast iteration, more natural body movement	Usage-based	Worth watching, less mature
Invideo AI	Best for script-to-finished-video	Subscription	Overkill for your use case

Don't use pre-rendered for daily calls. It feels broadcast, not relational. Reserve it for:

- Birthday messages, holiday greetings (where the family member could pre-approve the script).
- Medication reminders with the family face.
- Weekly summary videos delivered to the family dashboard.

Why I'm pushing you toward Tavus for the video layer

Three reasons:

1. **Consent verification is built in.** Tavus refuses to create a Replica without a specific consent recording from the person being cloned. This is a feature, not a limitation — it's your legal cover.
2. **BYO-LLM** means your prompt engineering, memory, and risk-detection logic stay in your control. HeyGen's Interactive Avatar pulls you toward their conversational platform.
3. **Sub-1s latency** is the threshold for "feels like a conversation" vs. "feels like a delay." Pre-rendered video can't get there; only real-time CVI platforms can.

What this costs at the unit level

Daily 5-minute voice call: ~\$0.10–0.15/call. Weekly 10-minute video call: ~\$1.50–3.00/call. Per user/month: voice (30 calls) + video (4 calls) ≈ **\$10–15/month in raw AI infra cost.**

This is why your pricing tier of \$39/month for Family is workable — you have 60%+ gross margin if you bundle video at weekly cadence.

5. Memory & Personalization Layer

The startup's real moat is here, not in the call quality.

Three tiers of memory

Tier	Storage	Refresh	Purpose
Within-call	In-process list of messages	Every turn	Coherence inside a single call

Tier	Storage	Refresh	Purpose
Recent context	Postgres summary table	After each call	"What did we talk about last week?"
Long-term semantic	pgvector embeddings	After each call	"She mentioned her grandson's name 4 months ago"

Post-call summary pipeline

After every call, run this in a Celery worker:

```

async def post_call_pipeline(call_id: str):
    transcript = await get_transcript(call_id)

    # 1. Generate a structured summary
    summary = await llm.chat(model="claude-haiku-4-5", messages=[{
        "role": "user",
        "content": f"""Summarize this call in the structured format below.
        Transcript: {transcript}

        Return JSON:
        {{
            "one_line_summary": "...",
            "mood": "happy|neutral|sad|anxious|confused",
            "topics_discussed": [...],
            "people_mentioned": [...],
            "places_mentioned": [...],
            "concerns_raised": [...],
            "physical_symptoms_mentioned": [...],
            "memorable_quotes": [...]
        }}"""])

    # 2. Generate embedding chunks for long-term memory
    chunks = chunk_transcript(transcript, max_tokens=300)
    for chunk in chunks:
        embedding = await openai.embeddings.create(
            model="text-embedding-3-large", input=chunk
        )
        await pgvector_insert(senior_id, chunk, embedding, call_id)

    # 3. Update behavioral baseline (see Section 7)
    await update_baseline(senior_id, transcript, summary)

    # 4. Compute risk score and decide alert level
    risk = await compute_risk(senior_id, transcript, summary)
    if risk.severity > threshold:
        await create_alert(senior_id, risk)

```

```
# 5. Persist everything
await save_summary(call_id, summary)
```

Memory retrieval at call start

```
async def build_call_context(senior_id: str):
    recent = await db.fetch_summaries(senior_id, n=3, order="desc")

    important_topics = await db.fetch(
        "SELECT topic, last_mentioned FROM recurring_topics "
        "WHERE senior_id=$1 ORDER BY importance_score DESC LIMIT 10",
        senior_id
    )

    today_context = await build_calendar_context(senior_id)
    # e.g., "Today is the anniversary of her husband's passing"
    #       "She has a doctor's appointment in 2 days"
    #       "Her grandson Aarav had his 5th birthday yesterday"

    return {
        "recent_summaries": recent,
        "important_topics": important_topics,
        "today_context": today_context,
    }
```

The reason this matters: a senior who says "remember I told you my back was hurting last week?" and gets a blank response loses faith instantly. The retrieval doesn't need to be sophisticated — it needs to be *reliable*.

6. Backend Infrastructure

Stack

Layer	Choice	Rationale
Web framework	FastAPI (Python 3.12)	Async-first, fits your AI/ML background
Database	Postgres 16 with pgvector	One database for relational + vector; simpler ops
Queue	Redis + Celery (or Dramatiq)	Post-call processing, scheduled calls
Real-time orchestration	LiveKit Agents	Skip 80% of voice-orchestration code
Auth	Clerk or Supabase Auth	Family-side only; seniors don't log in
LLM orchestration	LangGraph (not plain LangChain)	State machines for branching call flows
Telephony	Twilio (or Plivo for cost)	PSTN dial-out

Layer	Choice	Rationale
Video	Tavus CVI + LiveKit	Real-time avatar streaming
Frontend	Next.js 15 + shadcn/ui	Family dashboard only
Hosting	Railway (MVP) → AWS/GCP (scale)	Fast deploy, easy migration
Observability	PostHog (product) + Sentry (errors) + Logfire (LLM traces)	LLM tracing is non-optional
Secrets	Doppler or AWS Secrets Manager	Voice clones are biometric data — treat like crown jewels

Why LangGraph specifically

A senior conversation has branching states:

- Normal small talk
- Symptom mention (escalate to gentle probing)
- Emotional distress (slow down, more reflective)
- Confusion (very short responses, redirect to grounding)
- Explicit request to talk to family (live transfer)
- Emergency keywords (immediate family + emergency contact alert)

LangChain handles linear flows badly. LangGraph models these as states with transitions and is the right abstraction.

Project structure

```
wellring/
├── api/                # FastAPI routes (family dashboard backend)
├── workers/           # Celery: post-call processing, scheduling
├── voice/             # LiveKit agents, Twilio handlers
├── video/             # Tavus integration
├── llm/
│   ├── prompts/      # Versioned system prompts
│   ├── graphs/        # LangGraph state machines
│   └── memory/        # Retrieval logic
├── safety/           # Risk detection, alert routing
├── consent/          # Consent capture + vault
├── db/               # SQLAlchemy models, migrations (Alembic)
└── web/              # Next.js family dashboard
```

7. Risk Detection & Behavioral Baseline Engine

The original document called this the real moat. It's right. Generic sentiment analysis is useless; per-senior baselines are the product.

The two-layer approach

Layer 1: Universal rule-based signals. Things that always matter, regardless of baseline.

```
HARD_FLAGS = {
  "fall_mention": {"weight": 40, "patterns": ["i fell", "i slipped", "couldn't
get up"]},
  "chest_pain": {"weight": 50, "patterns": ["chest pain", "chest is tight"]},
  "suicidal_ideation": {"weight": 100, "patterns": [...]}, # immediate
escalation
  "medication_skipped": {"weight": 25, "patterns": ["didn't take my", "forgot my
pill"]},
  "wandering_disorientation": {"weight": 35, "patterns": ["don't know where",
"can't find my way"]},
}
```

Layer 2: Per-senior anomaly detection. Things that matter *relative to this person*.

For each senior, after ~30 calls you have a baseline distribution on:

- Average sentence length
- Words per minute
- Hesitation rate (filled pauses, "um", false starts)
- Vocabulary diversity (type-token ratio)
- Emotional valence (rolling 7-day mean)
- Topic distribution (do they always talk about gardening but now never?)
- Sleep mentions, appetite mentions, complaint frequency

The risk score is then a weighted combination:

```
risk_score = (
  sum(hard_flags_triggered_weights)
  + 30 * z_score(mood_today, baseline_mood)
  + 20 * z_score(hesitation_rate_today, baseline_hesitation)
  + 15 * z_score(vocab_diversity_today, baseline_vocab)
  - 10 * mentions_positive_event # protective factors
)
```

Alert routing

Score	Action
0–30	Daily summary in dashboard
31–60	Push notification to family
61–85	SMS + email to primary family contact
86–100	Phone call to family + push to secondary contacts

Score	Action
Hard flag: suicidal ideation, severe symptoms	Immediate call to family + locale-appropriate emergency resources surfaced to senior

Critical: never have the AI itself triage as a medical authority. The dashboard says "She mentioned chest tightness; we recommend you call her or her doctor." It does *not* say "Likely cardiac event."

8. Database Schema

```
-- Family unit
CREATE TABLE families (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  plan TEXT NOT NULL, -- 'basic', 'family', 'premium'
  created_at TIMESTAMPTZ DEFAULT NOW(),
  timezone TEXT NOT NULL
);

-- Family members (the paying users)
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  family_id UUID REFERENCES families(id),
  email TEXT UNIQUE NOT NULL,
  phone TEXT,
  name TEXT NOT NULL,
  role TEXT, -- 'primary_contact', 'secondary'
  voice_clone_id TEXT, -- ElevenLabs voice ID if they've cloned
  avatar_replica_id TEXT, -- Tavus replica ID
  consent_records JSONB, -- linked to consent_records table
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- The seniors (the people being called)
CREATE TABLE seniors (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  family_id UUID REFERENCES families(id),
  name TEXT NOT NULL,
  preferred_name TEXT, -- 'Mom', 'Dadi', 'Grandma Susan'
  phone TEXT NOT NULL,
  timezone TEXT NOT NULL,
  locale TEXT, -- 'en-IN', 'hi-IN', 'en-US'
  call_schedule JSONB, -- {time: '09:00', days: [...]}
  cloned_from_user_id UUID REFERENCES users(id), -- whose voice/face is used
  profile_summary TEXT, -- LLM-maintained summary of who they are
  baseline_metrics JSONB, -- computed behavioral baseline
  alert_thresholds JSONB,
  medications JSONB,
  doctor_info JSONB,
  disclosure_state JSONB, -- last_disclosed_at, calls_since_disclosure
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```

-- Consent records (legal cover; immutable)
CREATE TABLE consent_records (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id),
  senior_id UUID REFERENCES seniors(id),
  consent_type TEXT, -- 'voice_clone', 'likeness_clone', 'data_processing'
  scope JSONB, -- {use_for: [...], do_not_use_for: [...]}
  recording_url TEXT, -- the actual consent recording, encrypted-at-rest
  verification_phrase TEXT,
  granted_at TIMESTAMPTZ NOT NULL,
  revoked_at TIMESTAMPTZ, -- null = active
  legal_basis TEXT -- 'GDPR Art 6(1)(a)', 'DPDP s.6', etc.
);

-- Calls
CREATE TABLE calls (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  senior_id UUID REFERENCES seniors(id),
  scheduled_at TIMESTAMPTZ NOT NULL,
  started_at TIMESTAMPTZ,
  ended_at TIMESTAMPTZ,
  duration_seconds INT,
  channel TEXT, -- 'voice', 'video'
  status TEXT, -- 'scheduled', 'in_progress', 'completed', 'missed', 'failed'
  transcript_url TEXT, -- encrypted blob storage
  recording_url TEXT, -- only if consented
  summary JSONB,
  sentiment_score FLOAT,
  risk_score FLOAT,
  behavioral_deltas JSONB, -- per-metric z-scores vs baseline
  disclosure_given BOOLEAN DEFAULT FALSE
);

-- Long-term semantic memory
CREATE TABLE memory_chunks (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  senior_id UUID REFERENCES seniors(id),
  call_id UUID REFERENCES calls(id),
  content TEXT NOT NULL,
  embedding VECTOR(3072), -- text-embedding-3-large dimension
  tags TEXT[],
  importance_score FLOAT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX memory_chunks_embedding_idx ON memory_chunks
  USING hnsw (embedding vector_cosine_ops);

-- Recurring topics (cheap retrieval index)
CREATE TABLE recurring_topics (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  senior_id UUID REFERENCES seniors(id),
  topic TEXT NOT NULL,
  first_mentioned TIMESTAMPTZ,

```

```
        last_mentioned TIMESTAMPTZ,
        mention_count INT DEFAULT 1,
        importance_score FLOAT,
        sentiment_trend FLOAT[]
    );

-- Alerts
CREATE TABLE alerts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    senior_id UUID REFERENCES seniors(id),
    call_id UUID REFERENCES calls(id),
    severity TEXT, -- 'info', 'medium', 'high', 'critical'
    category TEXT, -- 'mood', 'physical', 'cognitive', 'medication', 'hard_flag'
    reason TEXT,
    triggered_signals JSONB,
    notified_users JSONB, -- which users got which channels
    acknowledged_by UUID REFERENCES users(id),
    acknowledged_at TIMESTAMPTZ,
    resolved_at TIMESTAMPTZ
);

-- Behavioral baseline (computed daily)
CREATE TABLE baselines (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    senior_id UUID REFERENCES seniors(id),
    computed_at TIMESTAMPTZ DEFAULT NOW(),
    sample_size INT, -- how many calls feed this baseline
    metrics JSONB, -- {mood: {mean, std}, wpm: {mean, std}, ...}
    is_active BOOLEAN DEFAULT TRUE
);
```

9. Compliance, Privacy & Legal

Jurisdiction matrix (your customers and data)

Region	Law	Key obligations
India (likely launch)	DPDP Act 2023 + IT Rules 2021	Explicit consent for biometric data (voice clones), data localization, breach notification
EU	GDPR + AI Act	Article 50 (synthetic media disclosure), DPIA required, lawful basis, right to erasure
US	State-level (CCPA, BIPA in IL, etc.)	Illinois BIPA is the strictest — written consent for biometric capture, \$1K–\$5K per violation
UK	UK GDPR + Online Safety Act	Similar to EU

Minimum compliance set for launch

1. **Consent vault** (the `consent_records` table above). Recorded, verifiable, revocable.
2. **Encryption at rest** for all transcripts, recordings, embeddings. Use AWS KMS or equivalent.
3. **Encryption in transit** — obviously, but also for internal service-to-service.
4. **Data retention policy**. Default: transcripts auto-purge after 90 days unless flagged. Voice clones retained only while consent is active.
5. **Right to deletion** — one-click full-erase from family dashboard. Must cascade to vector store.
6. **DPIA (Data Protection Impact Assessment)** before launching in EU.
7. **Don't position as medical**. Your terms of service must make this clear: WellRing is a wellness companion, not a medical device, not an emergency dispatch system. The original document was right about this.
8. **Mandatory reporter ambiguity**. In some jurisdictions, if your system detects suicidal ideation or abuse, you may have a legal duty to escalate to authorities. Get a lawyer to draft your protocol *before* the first such event happens, not after.

Avoid HIPAA at MVP stage

If you don't accept payments from insurers/Medicare, don't store medication dosages, and don't transmit data to providers, you can position as non-HIPAA covered. This dramatically reduces compliance burden for the first 12 months. Plan for HIPAA when you do B2B (senior living facilities, payor partnerships) — not before.

10. Tool Stack

Free-tier MVP build (≈ \$0–\$50/mo for first 10 families)

Category	Tool	Free tier
Telephony	Twilio	\$15 trial credit
STT	Deepgram Nova-3	\$200 credit (no card), no expiry
LLM	Anthropic / OpenAI / Google	\$5 free on most; alternatively use OpenRouter for routing
TTS / Voice Clone	ElevenLabs	Free tier: 10K credits, but NO commercial license. Move to Starter (\$5) immediately
Real-time orchestration	LiveKit Cloud	Free up to 50 connection-minutes/month
Video Avatar	Tavus	Free 25 minutes Replica + 100 conversational minutes/month on free tier
Database	Supabase	Free Postgres + pgvector up to 500MB
Queue	Upstash Redis	Free up to 10K commands/day
Hosting	Railway	\$5 starter credit; ~\$5/mo for hobby plan
Frontend	Vercel	Free hobby tier
Auth	Clerk	Free up to 10K MAU

Category	Tool	Free tier
Error tracking	Sentry	Free up to 5K events/mo
LLM observability	Logfire / Langfuse	Both have generous free tiers
Analytics	PostHog	Free up to 1M events/mo

You can validate with 5–10 real families for under \$50/month of infra.

Paid production stack ($\approx$ \$300–\$800/mo for 50 active families)

Category	Tool	Monthly cost
Twilio	PSTN (~30 min/family/mo $\times$ 50 families)	~\$25
Deepgram Growth or PAYG	~1500 min/mo	~\$12
ElevenLabs Pro	Cloning + TTS	\$99
Tavus	CVI usage (50 families $\times$ ~40min video/mo)	~\$200–400
LLM (Claude/GPT) with prompt caching	~10M input tokens/mo	~\$30–80
Embeddings	text-embedding-3-large	~\$5
Supabase Pro	DB + storage	\$25
Railway / Render	Compute	~\$50
Clerk Pro	If you exceed free	\$25
Sentry / Logfire / PostHog	Observability	~\$60
Total		~\$530–800

At \$39/family/month $\times$ 50 families = \$1,950 revenue. ~60–70% gross margin. This works.

11. Cost Model

Per-call breakdown (5-minute voice call)

Component	Cost
Twilio outbound (India mobile)	\$0.065
Deepgram STT (5 min streaming)	\$0.039
LLM (avg 5K input + 1K output tokens with prompt caching)	\$0.018
ElevenLabs TTS (Flash, ~5 min generated)	\$0.025
Post-call summarization + embeddings	\$0.003
Total per voice call	~\$0.15

Per-call breakdown (10-minute video call)

Component	Cost
Tavus CVI streaming (10 min)	~\$2.00
Underlying STT + LLM + TTS	~\$0.30
Post-call processing	\$0.005
Total per video call	~\$2.30

Per user per month (typical Family tier)

- 30 voice calls × \$0.15 = \$4.50
- 4 video calls × \$2.30 = \$9.20
- **Total raw cost: ~\$13.70/user/mo**

At \$39/mo pricing: ~65% gross margin. At \$19/mo Basic (voice-only): ~76% margin. These are healthy SaaS economics.

12. MVP Build Plan

A 45-day plan for a solo technical founder, validated against the assumption that you do AI/ML + Python well and use Lovable/v0 for frontend acceleration.

Week 1: Foundation

- Supabase project, schema migrations
- Clerk auth
- Twilio account + DLT registration (India)
- ElevenLabs + Deepgram + Anthropic accounts, API keys vaulted
- Skeleton FastAPI + Next.js repos
- LiveKit Agents hello-world

Week 2: Single-turn voice loop

- Twilio outbound call → Media Streams WebSocket
- Deepgram streaming integration
- Single LLM call without memory
- ElevenLabs TTS playback
- Goal: make a hardcoded call to your own phone where you can have a 3-minute conversation

Week 3: Conversation state + memory

- LangGraph state machine for the conversation
- pgvector setup
- Post-call summary pipeline (Celery)
- Memory retrieval at call start
- Goal: call your own mom (with consent!) — the AI remembers what you talked about yesterday

Week 4: Cloning + consent

- Voice cloning flow: family member records sample, ElevenLabs Pro clone created
- Consent capture: separate consent recording flow, stored in vault
- System prompt updated to use cloned voice + reference family relationship
- Disclosure scheduler
- Goal: call your mom in your own cloned voice; she knows what it is

Week 5: Family dashboard + risk detection

- Next.js dashboard: schedule, transcripts (redacted), summaries, alerts
- Risk scoring v1 (hard flags only; behavioral baseline needs more data)
- Alert routing (SMS via Twilio, email via Resend, push via OneSignal)
- Goal: 3 real families using it daily

Week 6: Video pilot

- Tavus integration: Replica creation flow, conversation session
- WebRTC frontend (LiveKit React components)
- Video-call scheduling option
- Goal: one weekly video call working end-to-end

Week 7 (buffer): Behavioral baseline, polish, observability

- After 4 weeks of real call data, compute first baselines
- LLM tracing in Langfuse
- Pricing pages + Stripe
- Goal: charge the first 5 families

Concierge stage before any of this

Spend 2 weeks before week 1 doing what the original document recommended: manually monitor 5 families. Call them yourself daily. Take notes. The instinct for what alerts matter, what tone works, what seniors actually say — none of this is in the technical spec, and you'll save 3 months of wrong architecture by doing it.

13. Startup Rating

Parameter	Score	Reasoning
Market size & demographics	9 / 10	1.4B+ people over 60 globally by 2030; India alone will have 230M+ seniors. Adult-child caregiver segment (your buyer) is large, growing, anxious, and underserved.
Problem severity	9 / 10	Loneliness as a mortality risk is well-established (Holt-Lunstad meta-analyses put it on par with smoking). Family guilt and burnout are universal. People will pay to reduce these.
Emotional pull / willingness to pay	9 / 10	This is a "guilt purchase," which is one of the strongest possible purchase drivers. Comparable categories (Calm, Headspace, BetterHelp, Care.com)

Parameter	Score	Reasoning
		all benefit from the same dynamic.
Technical feasibility (today, 2026)	8 / 10	Every component exists. Real-time voice agents are mature (Tavus, LiveKit, Deepgram, ElevenLabs). Real-time video avatars cleared the "uncanny valley" in 2024-25. The hard part is integration and reliability at scale, not invention.
Ethical / reputational risk	5 / 10	Massive <i>if</i> you go the deceptive impersonation route. Manageable (7/10) if you commit to disclosed Echo Mode. The downside scenarios (a senior in cognitive decline experiences confusion, an alert is missed before a fall, a press story about "AI fooling grandma") are real and need product design to defuse.
Regulatory risk	4 / 10	Voice cloning is the fastest-tightening regulatory area in AI right now. India DPDP, EU AI Act, US TAKE IT DOWN Act, IL BIPA, and a dozen state laws all converge on biometric consent and synthetic media disclosure. You'll need a fractional GC by month 6.
Defensibility / moat	7 / 10	The voice/video stack is not a moat (anyone can buy ElevenLabs + Tavus). The moats are: (a) per-senior behavioral baselines built over months, (b) trust + consent records, (c) the alert-tuning data showing what <i>actually</i> predicts incidents, (d) distribution into caregiver communities. (a) and (c) compound.
Distribution / go-to-market	7 / 10	Caregiver Facebook groups, r/AgingParents, TikTok storytelling all work. The viral hook ("I built an AI that calls my grandma every morning") is naturally there. The B2C CAC for this audience is reasonable — adult children search actively.
Unit economics	7 / 10	At \$19–39/mo with the cost model above, gross margin is 60–75%. Healthy. Watch out for video usage at the Premium tier — Tavus minutes are your highest variable cost.
Competitive landscape	6 / 10	Big incumbents (Best Buy Health/Lively, GreatCall, Papa Inc, Genie AI) are circling. Most are hardware-heavy or human-companion-marketplaces. AI-native pure-play is still relatively open, but window is closing — 18–24 months.
Founder fit (you, with AI/ML + Python)	8 / 10	The technical surface area maps to your strengths. Your weakness is likely sales/distribution and healthcare/legal — you'll need either a co-founder or fractional help here within 6 months.
Exit potential	8 / 10	Strategic buyers are real: AARP, Teladoc, Best Buy Health, Amazon Health, Hinge Health, Papa. AI elder care is in early M&A. Trade sale at \$50–200M in 4–6 years is plausible if you nail the consent/safety story.
Capital efficiency to PMF	7 / 10	Reachable PMF on ~\$200–500K. The infra costs and AI APIs are not the bottleneck; people are. One founder + one freelance designer + one fractional clinical advisor can get to first 500 paying families.

Overall rating

7.4 / 10 as a startup idea — strong, with two specific risks you must manage from day one:

1. **Don't build the deceptive version.** Echo Mode with disclosure is the only viable path. Bake disclosure into the product itself (every 10th call), not just the terms of service.
2. **Build the consent vault before the call orchestrator.** The temptation as a technical founder is to build the cool AI loop first and bolt on legal cover later. Reverse this. Your defensibility, vendor compliance, and exit readiness all depend on having an unimpeachable record of who consented to what.

If you do those two things, this idea is in the top decile of AI-SaaS opportunities I'd evaluate right now. The technology has arrived, the demographic wave is real, and the willingness to pay is unusually high for consumer software.

Document version: 1.0 · Prepared as build-ready engineering reference. Treat pricing and vendor capability claims as of May 2026; re-verify before commercial commitments.